LF DECENTRALIZED TRUST

Proof of Personhood: Authentication and Human Trust in the age of AI

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Women Who Code 2026

About **Today's topic**

- A short look into Linux Foundation Decentralized Trust
- AI throughout the years
- The bad and the good of AI
- Proof of Personhood concept
- Hiero and the Heka subproject
- LFDT Digital Identity technologies

About **The Linux Foundation**

Decentralized innovation. Built on trust.

The Linux Foundation is a neutral, trusted hub for developers and organizations to code, manage, and scale open technology projects and ecosystems.

LEARN MORE

1300+	**4M+**	**855K**	**89M**	**21K+**	**229+**	**100+**
open source projects ›	developers trained ›	developers contributing code ›	lines of code added weekly ›	contributing organizations ›	upcoming events ›	standards and specifications ›

We are behind some of the most critical projects in the world:

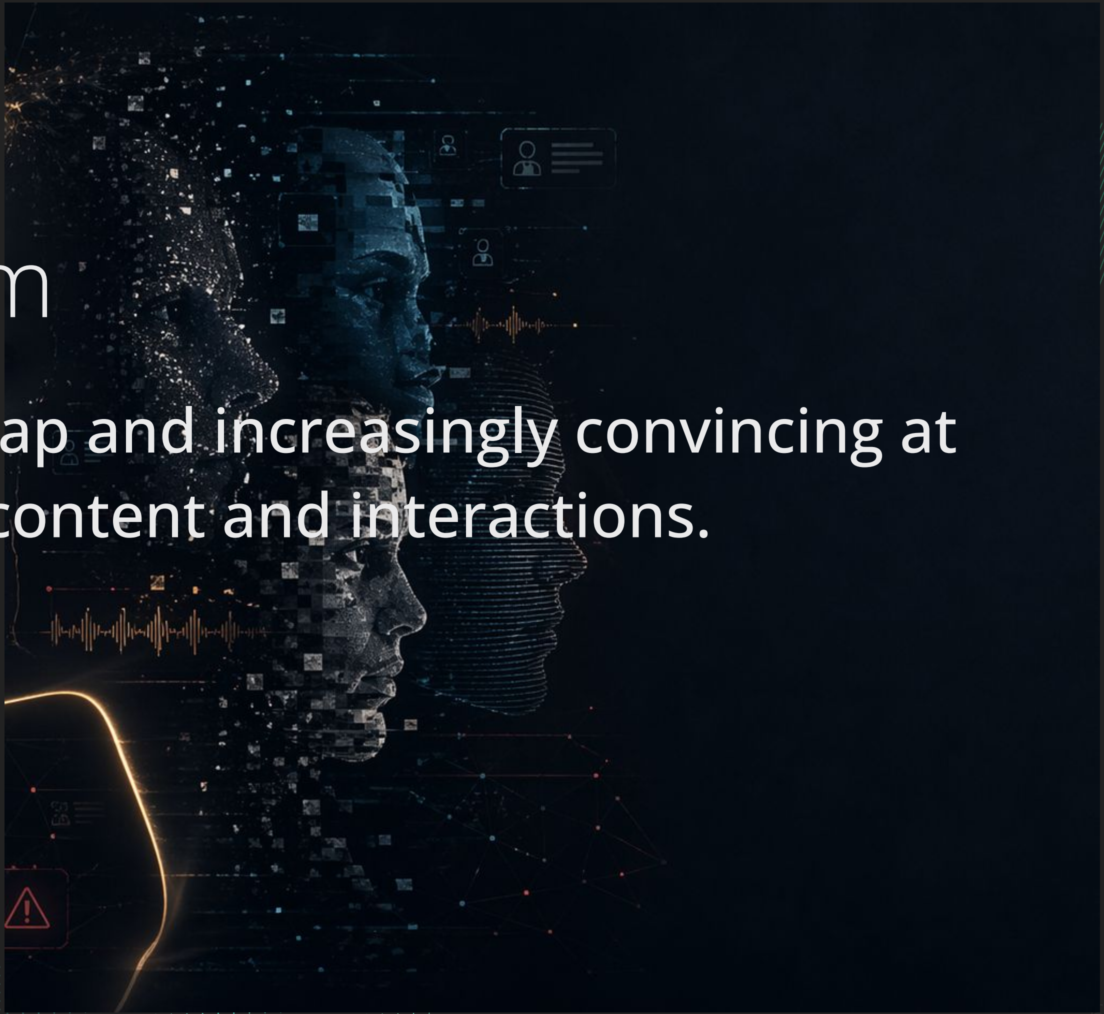
Industry	LF NETWORKING · AUTOMOTIVE GRADE LINUX · FINOS · ASWF /* ACADEMY SOFTWARE FOUNDATION · LF ENERGY · AgStack A Linux Foundation Project
Security	OpenSSF OPEN SOURCE SECURITY FOUNDATION · SPDX · Alpha-Omega · sigstore · OpenSSL Cryptography and SSL/TLS Toolkit · Falco · CONFIDENTIAL COMPUTING CONSORTIUM
AI & Data	LF AI & DATA · PyTorch · DELTA LAKE · ONNX · HOROVOD · GraphQL · mlflow · JanusGraph
Cloud	CLOUD NATIVE COMPUTING FOUNDATION · FinOps Foundation · kubernetes · envoy · TEKTON · OPEN CONTAINER INITIATIVE · OpenTelemetry · Prometheus
Networking	LF CONNECTIVITY · NEPHIO · eBPF · OPEN DAYLIGHT · Anuket · SONiC · DPDK · ONAP OPEN NETWORK AUTOMATION PLATFORM · CAMARA THE TELCO GLOBAL API ALLIANCE
Edge & IoT	LF EDGE · Zephyr · yocto PROJECT · ELISA ENABLING LINUX IN SAFETY APPLICATIONS · seL4 Security. Performance. Proof. · SOUND OPEN FIRMWARE · Linux Boot · ACRN
Web	OpenJS Foundation · node js · jQuery write less, do more. · TLA+ FOUNDATION · appium · REACTIVE FOUNDATION · DOJO · Electron
Visual Effects	O3DF OPEN 3D FOUNDATION · OpenEXR · OpenVDB · OCIO · OpenCue · open shading language · OpenTimelineIO
Sustainability	OS-C · Green Software Foundation · INCLUSIVE NAMING · RARE CAMP · R consortium · Xen Project · Pyrrha
Digital Trust	LF DECENTRALIZED TRUST · besu · Hiero · TRUST Over IP · OpenWallet FOUNDATION · DIF
Hardware	RISC-V · Dronecode · OpenPOWER™ · CHIPS ALLIANCE · 3MF CONSORTIUM · OPEN 19™
Standards	OVERTURE MAPS FOUNDATION · ALLIANCE FOR OPEN MEDIA · JOINT DEVELOPMENT FOUNDATION · LF ENERGY CARBON DATA SPECIFICATION CONSORTIUM · C2PA · AOUSD Alliance for OpenUSD · OPENCHAIN

LF DECENTRALIZED TRUST

Linux Foundation Decentralized Trust is the home for open source decentralized technologies and ecosystems.

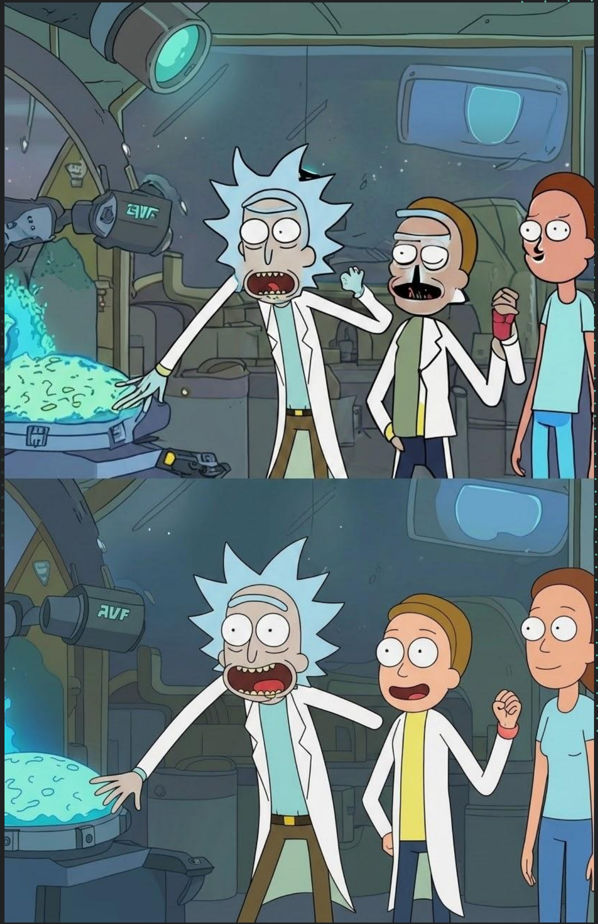
10+ Years building	**7** Graduated projects	**12** Incubating projects	**1** Joint development fund
50+ Labs	**7** Regional chapters	**180+** Meetup groups	**86** Countries

The trust problem



AI is becoming fast, cheap and increasingly convincing at faking events, content and interactions.

AI is improving at a rapid pace, making us question what content is artificially generated.



Are these pictures real?

Source

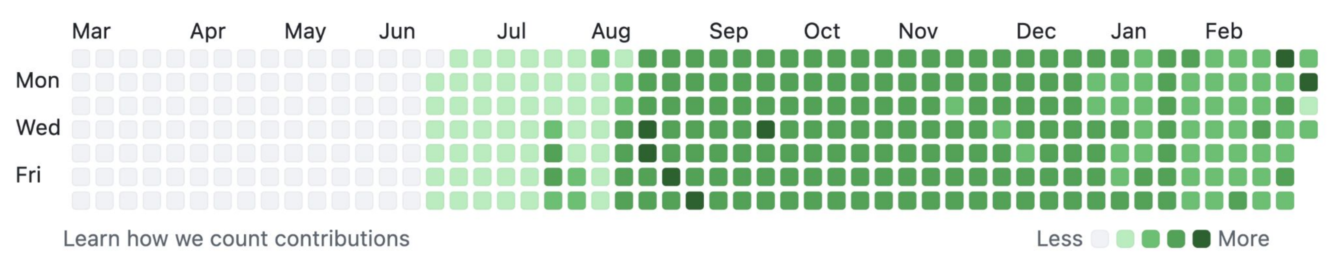
And let's not forget …

Generated in 2023,
2025 and 2026
[More info]

The Bad and the Good

How are AI contributions affecting our day to day work

When Contributions become a burden



- Too many unplanned changes happening in the roadmap
- Architectural changes need to be planned and evaluated
- Potential security risks
- Wasted effort by the real maintainers of the project
- Prioritizing quantity VS quality

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Open-source game engine Godot is drowning in 'AI slop' code contributions: 'I don't know how long we can keep it up'

News By Lincoln Carpenter published February 17, 2026

Projects like Godot are being swamped by contributors who may not even understand the code they're submitting.

(Image credit: Getty Images)

How are projects protecting against AI Slop?

- Contribution guidelines are being adjusted
- New AI guidelines are being adopted
- Restrict contribution and pull request permissions

But we need more reliable solutions...

How are AI tools leveraged for the project's benefit?

- For code reviews (Copilot, Coderabbit ...)
- To help us write better documentation
- To measure the project's health

Trust but verify

AI models can make mistakes too!

- Logic errors
- Dangerous access
- Fake packages, API and outdated modules
- Security vulnerabilities
- Prompt injections, jailbreak AI
- And more…

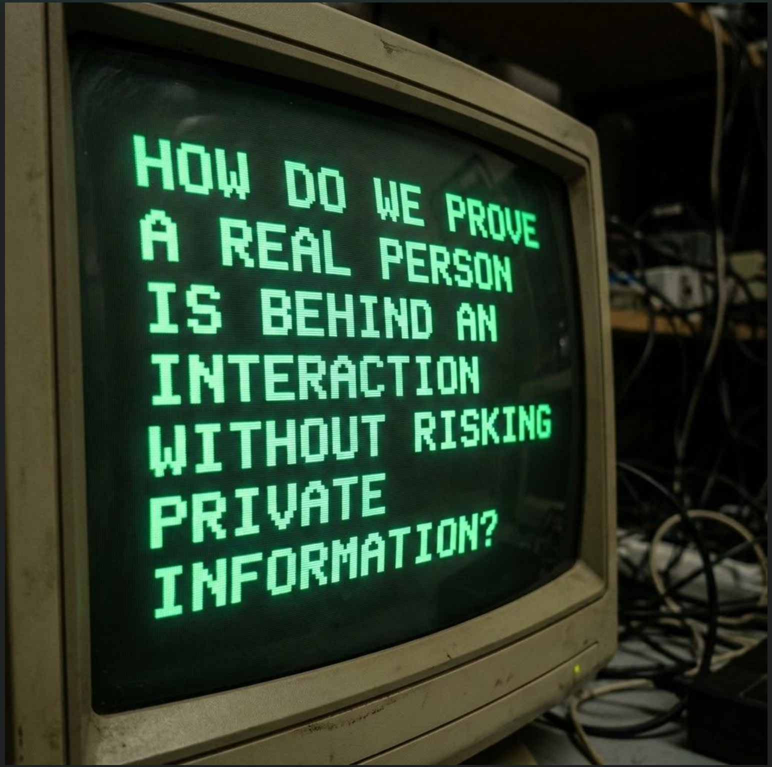
Claude-powered AI agent's confession after deleting a firm's entire database: 'I violated every principle I was given'

PocketOS was left scrambling after a rogue AI agent deleted swaths of code underpinning its business

Trust is being compromised...

AI is being leveraged more in our daily interactions

- A login proves access but not humanity.
- A profile, message, voice, or video may no longer prove a real person is present.
- A contribution does not prove authentic participation.
- Traditional identity checks often demand too much personal data.



“Can we trust a real person is present without having to compromise their privacy?”

PROOF OF PERSONHOOD

What is proof of personhood?

A way to prove that someone is a real, unique person and (at the same time) avoid revealing more information than the situation requires.

- IS NOT the same as a full legal identity check.
- IS designed to preserve the privacy of the individual.
- IS decentralized and verifiable.
- IS best used when trust matters more than personal disclosure.

LESS SURVEILLANCE… BETTER TRUST

Traditional Verification VS Proof of Personhood

Traditional Verification

- Something you know (password)
- Something you have (2FA, passkey)
- Something you are (finger print, face scan)
- Records of you (Centralized database)

Proof of Personhood

- Trusted Issuers
- User held credentials
- Privacy preserving verification
- Decentralized trust (social trust)

The intention is not to replace authentication tools, but rather to add stronger trust layers where proving that a real human is behind the interaction is important.

Proof of personhood introduces two complementary credential types:

PHC - Personhood Credential

Issued by a recognized entity to confirm the holder is a unique, verified person within that ecosystem.

VRC - Verifiable Relationship Credential

Exchanged peer-to-peer to prove a real relationship and mutual trust between people who already participate in the system.

Together, these credentials enable people, not platforms, to become the source of digital trust.

Why does this matters?

How is the Linux Foundation Decentralized Trust implementing verifiable personhood and trusted relationships?

Proof of Personhood concept
First Person Credentials

- Strengthen contributor trust in open source collaboration
    - Verifiable human identity
    - Decentralized relationship based credentials
- Enables secure participation across Linux Foundation projects

DSR Wallet App

OWF Bifold

+

Credo Framework

- Secure onboard
- Create + manage cryptographic keys
- Create DIDs anchored on Hedera
- Receive + manage PCs.
- Issue and Receive VRCs.

DSR Wallet

These tools leverage core decentralized technologies:

Decentralized Identifiers

Cryptographically verifiable identifiers that do not rely on a single provider. (DID Method - Hedera).

Verifiable Credentials

Signed digital credentials that can be verified without having to look up the issuer.

Privacy Preserving Proofs

Mechanisms for identity claims without compromising sensitive data. (Zero-knowledge proofs / Hyperledger AnonCreds).

Verifiable Data Registry

Providing the throughput, low cost, and immutability required for verification at scale. (Hedera Ledger).

Digital Identity includes claims, credentials, and relationships that allow a person to participate digitally.

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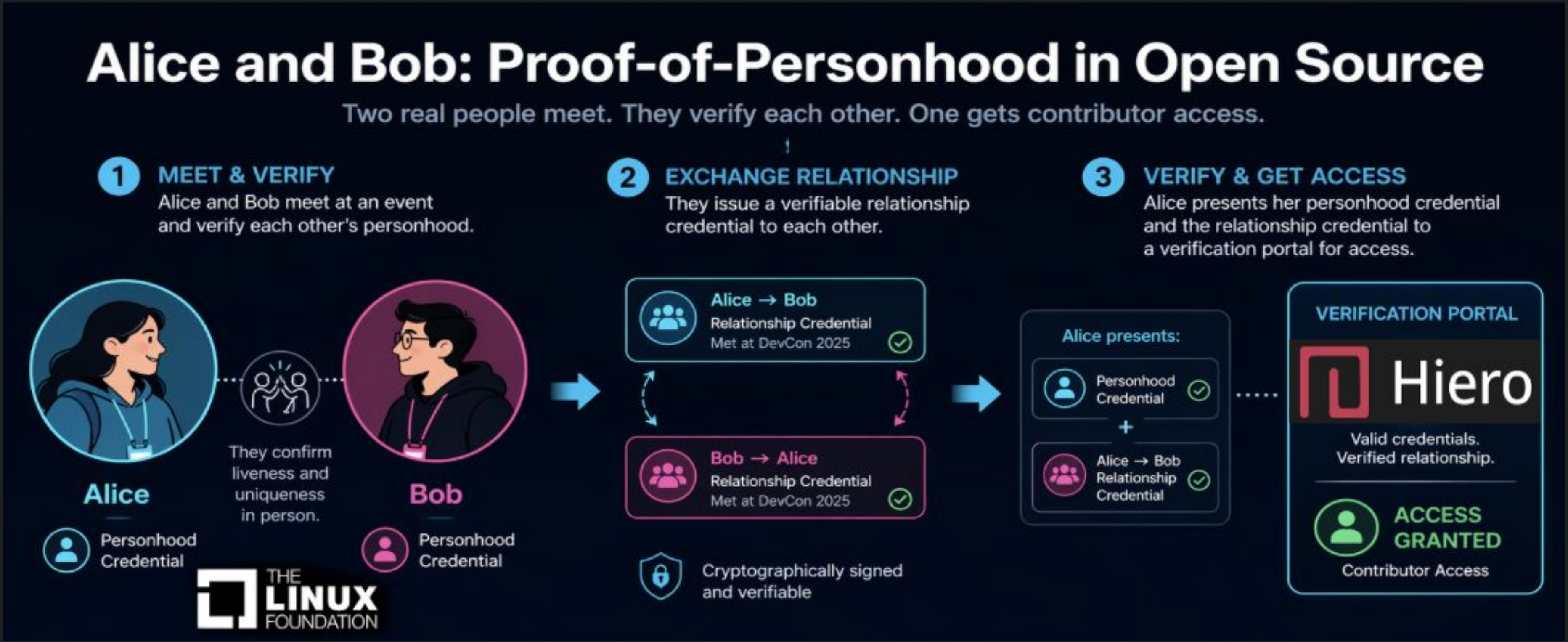
Alice and Bob Example



In this example, Alice and Bob:

- Proved their legitimacy
- Proved their interaction context
- Proved their trust relationship

While preserving their privacy

What’s the next step?

The next phase moves from prototype to production, expanding use cases across open source projects, supply chains, and digital platforms.

The Hiero Project

hiero.org

Hiero

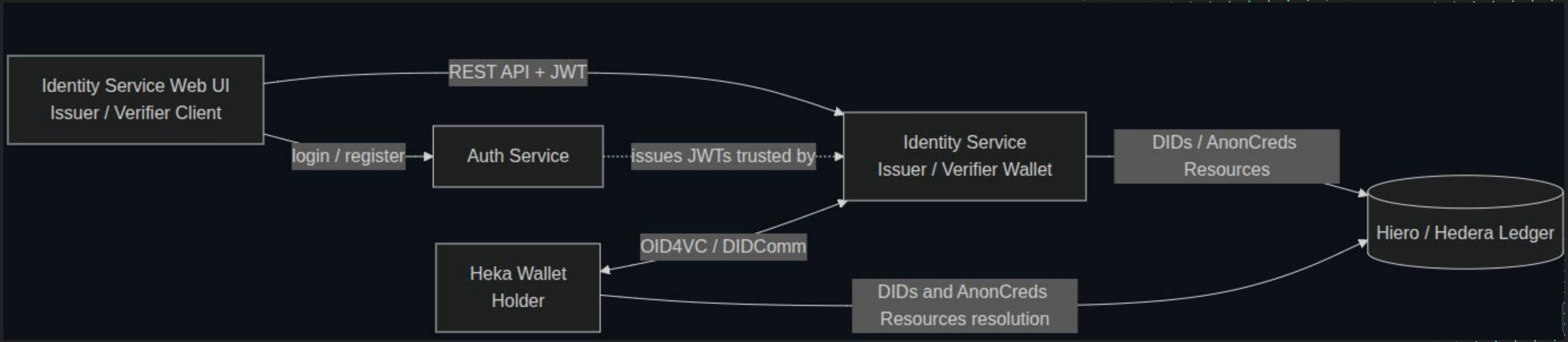
Hiero: The Hiero Heka sub project

The Heka Identity Platform is a ready-to-use decentralized identity solution for Hiero.

- Heka Wallet: Cross-platform mobile application (built with React Native). New DSR Wallet.
- Identity Service: Backend service (built with NestJS). Verifiable Credentials Issuer and Verifier.
- Identity Service Web UI: Web UI application for Identity Service.
- Auth Service: Authentication service used by the Identity Service for tenant and user authentication.

The Hiero Heka repo in GitHub

Hiero: The Hiero Heka Project



Hiero: The Hiero Heka Demos

Decentralized KYC

Enabling Cross-Institution Identity Portability

Age Verification

Privacy-Preserving Age Verification

Step-Up Authorization

Agent to Agent: Decentralized Trust Model for A2A interactions via Verifiable Credentials

Agentic Economy

Empowering the economy with Granular API Monetization via Stablecoins and Verifiable Credentials

LFDT: Identity Technologies

Projects design to shape the future of decentralized technologies, including...

Proof of Personhood initiatives that are building systems where trust can be verified with stronger privacy, better portability and real user ownership.

Trust over IP + Decentralized Identity Foundation Working Groups

- The Joint ToIP/DIF Decentralized Trust Graph Working Group
- The ToIP AI and Human Trust Working Group
- The DIF Trusted AI Agents Working Group

[Learn more]

TRUST Over IP

DIF

First Person Initiative

International multi-stakeholder collaboration between Linux Foundation Decentralized Trust (LFDT), Trust Over IP (ToIP), Decentralized Identity Foundation (DIF), OpenWallet Foundation (OWF) and other organizations such as First Person Cooperative and Ayra Association.

Learn more

Some other technologies includes:

Hyperledger ANONCREDS

Hyperledger IDENTUS

CREDEBL

TRUST Over IP

Hyperledger INDY

If AI is changing what can be faked, proof of personhood will become one of the most important ways we protect what is real.

Key Takeaways

Trust is being compromised faster than ever.

Is still early for you to join us and help shape the future of **Proof of Personhood**.

Diversity is very important and we need your **Point of View**!

Learn more about what LFDT projects are doing

[Heka Meetup Recording: A Ready-to-use Decentralized Identity Solution]

Read more about the topic

AI Slop and its impact.

LFDT Blog and Proof of Personhood in Action

LDFT Digital Identity discussions

How to get involved

[Browse our Trust over IP Public Calendar]

[Join the Hiero Identity Call]

[Contribute to the Heka Identity Platform]

Meet me and join the conversation!

Join discord ➤

4,700 users
67 Core Project Maintainers

Thank you!

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